School of Electrical Engineering, MIT Manipal

(ECE 3222) System on Chip Design – Flexible Core: Assignments 1-4, Marks = 20 (5x4)

Project Title: Design of Hardware IP for Image analysis & its FPGA Implementation

Motivation: Image processing hardware will play a critical role in the coming years as visual data becomes central to intelligent decision-making across diverse domains. Applications such as autonomous vehicles and drones require ultra-low latency and deterministic processing for real-time perception and navigation, which software-only solutions cannot reliably guarantee. Similarly, the growth of IoT and edge devices demands on-device image analysis to reduce power consumption, latency, and data transmission while preserving privacy. In satellite and remote sensing systems, hardware-accelerated image processing is essential for real-time enhancement, compression, and feature extraction from high-resolution imagery. Furthermore, biometric and biomedical image analysis relies on fast, accurate, and secure processing for tasks such as identity verification and medical diagnostics. Hardware-based image processing, particularly using FPGA and custom IPs, enables parallelism, efficiency, and reliability, making it a foundational technology for next-generation intelligent and autonomous systems.

Students are encouraged to explore classical image processing, AI/ML-inspired techniques, or optimization-based algorithms, provided the core computation is implemented as a hardware IP on FPGA. Projects that demonstrate real-time or near real-time processing using camera input or video streams will be highly appreciated and awarded additional credit.

Assignment 1: Datapath and Control Path Design **(5 marks)**

Assignment 2: RTL coding and Functional Simulation **(5 marks)**

Assignment 3: FPGA Implementation and Testing **(5 marks)**

Assignment 4: Final Report, Presentation and Demonstration **(5 marks)**

Project Objectives

By the end of this project, students should be able to:

- ✓ Understand and model image processing algorithms at hardware architecture level
- ✓ Design efficient datapath and control-path architectures
- ✓ Implement image processing algorithms using RTL (Verilog/VHDL/SystemVerilog
- ✓ Integrate the designed IP with FPGA peripherals
- ✓ Evaluate performance in terms of latency, throughput, resource utilization, and power
- ✓ Demonstrate a working FPGA-based image processing system

Scope and Flexibility

Any image processing or image analysis task is allowed
Input may be stored images, video frames, or live camera feed
AI/ML-based approaches are allowed only if mapped to hardware
The project must result in a reusable Hardware IP block

School of Electrical Engineering, MIT Manipal

(ECE 3222) System on Chip Design – Flexible Core: Assignments 1-4, Marks = 20 (5x4)

Assignments / Phases

Assignment 1: Datapath and Control Path Design

Algorithm description and hardware mapping + Literature/ Research Paper review
Datapath and control-path diagrams
FSM/state diagram
Fixed-point and bit-width analysis

Assignment 2: RTL Coding and Functional Simulation

RTL implementation of the IP + Literature/ Research Paper review
Testbench development
Functional simulation and waveform verification

Assignment 3: FPGA Implementation and Testing

FPGA synthesis and implementation
Timing and resource utilization analysis
On-board validation (camera/display optional)

Assignment 4: Final Report, Presentation and Demonstration

Final project report
Presentation slides
Live FPGA demonstration

Evaluation Criteria

Hardware-centric design
Correctness and efficiency
FPGA implementation quality
Innovation and documentation quality

Team Size: Minimum of 1 to a Maximum of 4 members are allowed per team.

School of Electrical Engineering, MIT Manipal

(ECE 3222) System on Chip Design – Flexible Core: Assignments 1-4, Marks = 20 (5x4)

Example Project Ideas (Indicative, Not Restrictive)

Optimization & Intelligent Techniques

- K-Means based Image Segmentation
- Genetic Algorithm based Image Segmentation
- PSO (Particle Swarm Optimization) based Image Segmentation
- Adaptive Thresholding ideas using ML-inspired logic

AI / ML on FPGA (Lightweight)

- Object Detection using simplified CNN or classical ML
- AI-based Edge Detection
- Image Classification using small neural networks

Classical / Algorithmic Image Processing

- Edge Detection (Sobel, Prewitt, Canny, or learned variants)
- Image Enhancement (contrast stretching, histogram equalization)
- Image Compression (DCT-based, run-length encoding, predictive coding)
- Image Deblurring or Noise Reduction
- Feature Extraction (corners, edges, gradients)

Hardware Accelerators

- Systolic array based
- Graph Neural Network based
- CNN Acceleration